

Kexin Chu

ML Systems Researcher

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SUMMARY

- Ph.D. candidate at UConn (Advisor: Prof. Wei Zhang) building ML infrastructure for deterministic inference, MoE acceleration, and privacy-aware LLM serving.
- 4+ years leading engineering teams (6–12 people) across ML-powered products, large-scale search recommendation systems, and production inference services handling hundreds of millions of daily requests.
- 9 publications in ML systems; research spans KV-cache sharing, vector databases for multimodal RAG, MoE inference, and dependable AI systems.

EXPERIENCE

QuarterMill Remote
Technical Lead Jun 2026 – Present

- Led a 6-person engineering team to design and build QuarterMill’s core product from scratch, owning architecture decisions, sprint planning, and delivery milestones.
- Built the data infrastructure layer on Nebula, PGVector, and PostgreSQL; integrated OCR/HTR pipelines to extract structured information from heterogeneous document sources.
- Developed agent-based workflows for automated quality assessment and claims extraction, powering the product’s primary value pipeline.

Dongchedi (ByteDance) Shanghai, China / Remote
LLM Algorithm Researcher Intern Apr 2025 – Oct 2025

- Worked on model-deployment optimization for AI Car Selection, supporting inference services handling 400K+ daily requests.
- Built a quantization, offload, and prefetch pipeline for models that could not fit on a single 8xH20 node.
- Addressed cross-node inference bottlenecks caused by memory pressure and interconnect data movement in large-model deployment.

Baidu, Inc. Beijing, China
Senior Software Engineer / Architecture Lead Jul 2020 – Aug 2024

- Led a 12-person team as Search Recommendation architecture owner; promoted from T3 to T5 through sustained technical leadership across Push & Recommendation, DeepQA Search, streaming indexing, and LLM access control.
- Designed offline/online recommendation pipelines (Spark, Hadoop for batch; Golang + Gin for serving); cut data freshness latency from 24h to 30min and incremental indexing from 48h to 5min, supporting 600M pushes/day and 150M daily visits.
- Refactored monolithic PHP services into modular Go services, reducing long-tail latency by 63% and improving observability and parallel retrieval efficiency for high-traffic search workloads.
- Built access-control infrastructure for Baidu’s LLM-powered search features, enabling secure rollout and supporting 2M+ user registrations after launch.

RESEARCH PUBLICATIONS (* = REPRESENTATIVE WORK)

- LLM Memory Systems.**
- MCaM* (ICDCS 2025)*: Targets repeated-prefix prefill overhead in shared LLM serving by coordinating KV-cache reuse across HBM, DRAM, and SSD tiers. It cuts prefill latency by 69% and raises throughput by 3.3x.
 - SafeKV* (arXiv 2025)*: Targets privacy-safe shared KV-cache reuse in multi-tenant serving by isolating sensitive blocks while preserving reuse for non-sensitive prefixes. It preserves privacy while improving throughput by 2.66x over strict isolation.
 - ReLACE* (under review)*: Targets recall drift in dynamic multimodal retrieval under OOD queries and online updates to multimodal data. It maintains 93.7%–97.8% recall across changing workload conditions.
 - SQLens* (ongoing): Targets filtered vector search inefficiency when SQL predicates are applied after ANN traversal in PostgreSQL-based multimodal RAG backends. It delivers 1.4x–4.2x speedups across different selectivity settings.

- LLM Inference Optimization.**
- ExpertFlow* (arXiv 2025): Targets host–GPU expert swap stalls in memory-constrained MoE inference through adaptive prefetching and cache-aware expert routing. It reduces model stall time to less than 0.1% of baseline.
 - DynaExq* (arXiv 2025): Targets precision misallocation under strict GPU memory budgets in MoE serving by dynamically promoting hot experts instead of using a fixed bit-width map. It delivers up to 2.73x higher throughput over ExpertFlow.
 - MarginGate* (arXiv 2026): Targets batch-induced token flips in BF16 decoding without always-on verification. It restores 100% deterministic decoding with 15.05%–18.56% trigger rates and cuts LLM-42 latency overhead by up to 2.23x.
 - LQM-ContextRoute* (arXiv 2026): Targets same-function tool-provider routing under runtime load and quality heterogeneity in LLM agents. It improves search F1 by 2.18 points and StrategyQA accuracy by up to 18 points over SW-UCB.

EDUCATION

University of Connecticut, CT, USA	Ph.D. in Computer Science and Engineering, GPA 3.91	Aug 2024 – Present
Hefei University of Technology, China	M.S. & B.S. in Electrical Engineering	Aug 2013 – Jun 2020

TECHNICAL SKILLS

Languages	Python (proficient), C++ (proficient), Go (familiar), SQL (familiar)
ML / Serving	PyTorch, Transformers, vLLM, SGLang, CUDA
Systems / Data	Linux, FastAPI, Redis, PostgreSQL, PGVector, Faiss, DiskANN, Docker, Git, Kafka, gRPC

AWARDS

Predoctoral Fellowship, University of Connecticut (2025, 2026); Baidu Pride Special Award (2022); Baidu Breakthrough Innovation Award (2021–2022); National Scholarship, Hefei University of Technology (2018, 2019)